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N + - Network Fundamentals & Building Networks.

Inter-VLAN Routing.

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TASK 1.

❖ On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.

- **VLAN Configuration:**

enable
configure terminal

vlan 10
name Insta
vlan 20
name Zomato
vlan 30
name Flipkart

Assign PCs to VLANs:

interface fa0/1
switchport mode access
switchport access vlan 10
exit

interface fa0/2
switchport mode access
switchport access vlan 20
exit

interface fa0/3
switchport mode access
switchport access vlan 30
exit

Verify Configuration:

show vlan brief

Result: Three VLANs Insta, Zomato, and Flipkart were successfully created, with one PC assigned to each VLAN.

TASK 2.

❖ Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., GigabitEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.

- **Configure Router on a Stick:**

Router Configuration:

enable
configure terminal

interface g0/0
no shutdown
exit

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit

VLAN Gateway Addresses:

- Insta (VLAN 10): 192.168.10.1
- Zomato (VLAN 20): 192.168.20.1
- Flipkart (VLAN 30): 192.168.30.1

Result: Three router sub-interfaces were configured successfully, providing a gateway for each VLAN.

TASK 3.

❖ Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command, Use the switchport mode trunk command on the relevant interface.

- **Configure Trunk Port on the Switch:**

enable
configure terminal

interface fa0/4
switchport mode trunk
no shutdown
exit

Verify Trunking:

show interfaces trunk

Result: The switch port connected to the router is configured as a trunk port, allowing traffic from VLANs 10, 20, and 30.

TASK 4.

❖ Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.

- **Test Inter-VLAN Connectivity:**

IP Address Configuration:

- Insta PC: 192.168.10.2/24, Gateway: 192.168.10.1
- Zomato PC: 192.168.20.2/24, Gateway: 192.168.20.1

Ping Test
From the Insta PC, open Command Prompt and enter:

ping 192.168.20.2

Result: The ping succeeds because Router on a Stick is configured with sub-interfaces for both VLANs, allowing communication between Insta and Zomato.

TASK 5.

❖ Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.

- **Add a New VLAN Spotify:**

Switch Configuration:

enable
configure terminal

vlan 40
name Spotify
exit

interface fa0/4
switchport mode access
switchport access vlan 40
exit

interface fa0/5
switchport mode trunk
exit

Router Configuration:

enable
configure terminal

interface g0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
exit

Spotify PC Configuration:

- IP Address: 192.168.40.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.40.1

Result: Spotify VLAN 40 is added with its own router sub-interface, while the existing VLANs remain configured.